



What is probability?

PROBABILITY IS THE LIKELIHOOD OF SOMETHING HAPPENING.

Math can be used to calculate the likelihood or chance that something will happen.

How is probability shown?

Probabilities are given a value between 0, which is impossible, and 1, which is certain. To calculate these values, fractions are used. Follow the steps to find out how to calculate the probability of an event happening and then how to show it as a fraction.

▷ Total chances

Decide what the total number of possibly outcomes is. In this example, with 5 candies to pick 1 candy from, the total is 5—any one of 5 candies may be picked.



there are 5 candies, 4 are red and 1 is yellow

▷ Chance of red candy

Of the 5 candies, 4 are red. This means that there are 4 chances out of 5 that the candy chosen is red. This probability can be written as the fraction $\frac{4}{5}$.

$$\frac{4}{5}$$

total number of red candies that can be chosen

total of 5 candies to choose from

▷ Chance of yellow candy

Because 1 candy is yellow there is 1 chance in 5 of the candy picked being yellow. This probability can be written as the fraction $\frac{1}{5}$.

$$\frac{1}{5}$$

1 yellow candy can be chosen

total of 5 candies to choose from

total of specific events that can happen

$$\frac{1}{8}$$

total of all possible events that can happen

◁ Writing a probability

The top number shows the chances of a specific event, while the bottom number shows the total chances of all of the possible events happening.

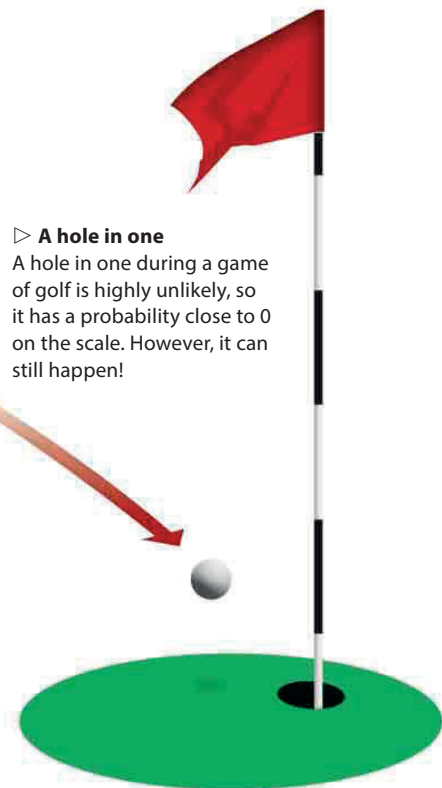


△ Identical snowflakes

Every snowflake is unique and the chance that there can be two identical snowflakes is 0 on the scale, or impossible.

0

IMPOSSIBLE



▷ A hole in one

A hole in one during a game of golf is highly unlikely, so it has a probability close to 0 on the scale. However, it can still happen!

UNLIKELY

▷ Probability scale

All probabilities can be shown on a line known as a probability scale. The more likely something is to occur the further to the right, or towards 1, it is placed.

LESS LIKELY

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◀ 64–65 Converting fractions, decimals, and percentages

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